Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.5903748043654392, median 0.3925340724012978, std: 0.5799936052648395
Gyroscope error (imu0): mean 0.6305567749744811, median 0.3433229804704081, std: 0.8499501180389034
Accelerometer error (imu0): mean 0.7077882463583302, median 0.40067041207757376, std: 1.0307635044302197

Residuals

Reprojection error (cam0) [px]: mean 0.5903748043654392, median 0.3925340724012978, std: 0.5799936052648395
Gyroscope error (imu0) [rad/s]: mean 0.02489699573098075, median 0.013555814667862804, std: 0.033359554508343144
Accelerometer error (imu0) [m/s^2]: mean 0.2747973181243135, median 0.15555945617517244, std: 0.40019179195926136

Transformation (cam0):

T_ci: (imu0 to cam0):
[[0.06964907 -0.99394843 0.08494427 -0.03709362]
 [0.31225411 -0.05915011 -0.94815539 -0.04677745]
 [0.94744202 0.09256234 0.30624473 0.04847292]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[0.06964907 0.31225411 0.94744202 -0.02873529]
 [-0.99394843 -0.05915011 0.09256234 -0.0441228]
 [0.08494427 -0.94815539 0.30624473 -0.05604598]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.005571981714868694

Gravity vector in target coords: [m/s^2]
[-0.57707962 -9.61979608 1.81519293]

Calibration configuration
==========

cam0

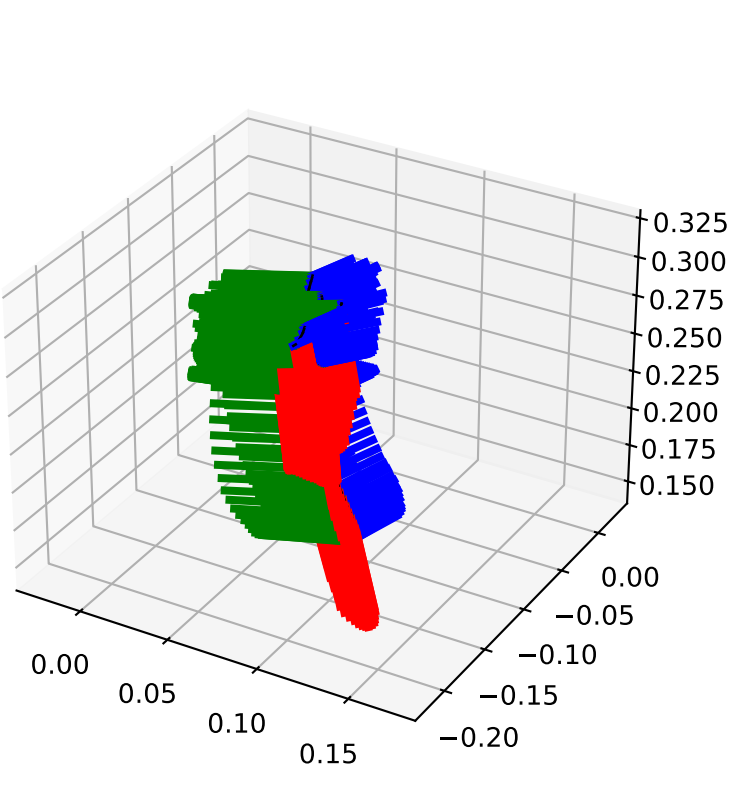
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.02 [m]
    Spacing 0.006 [m]

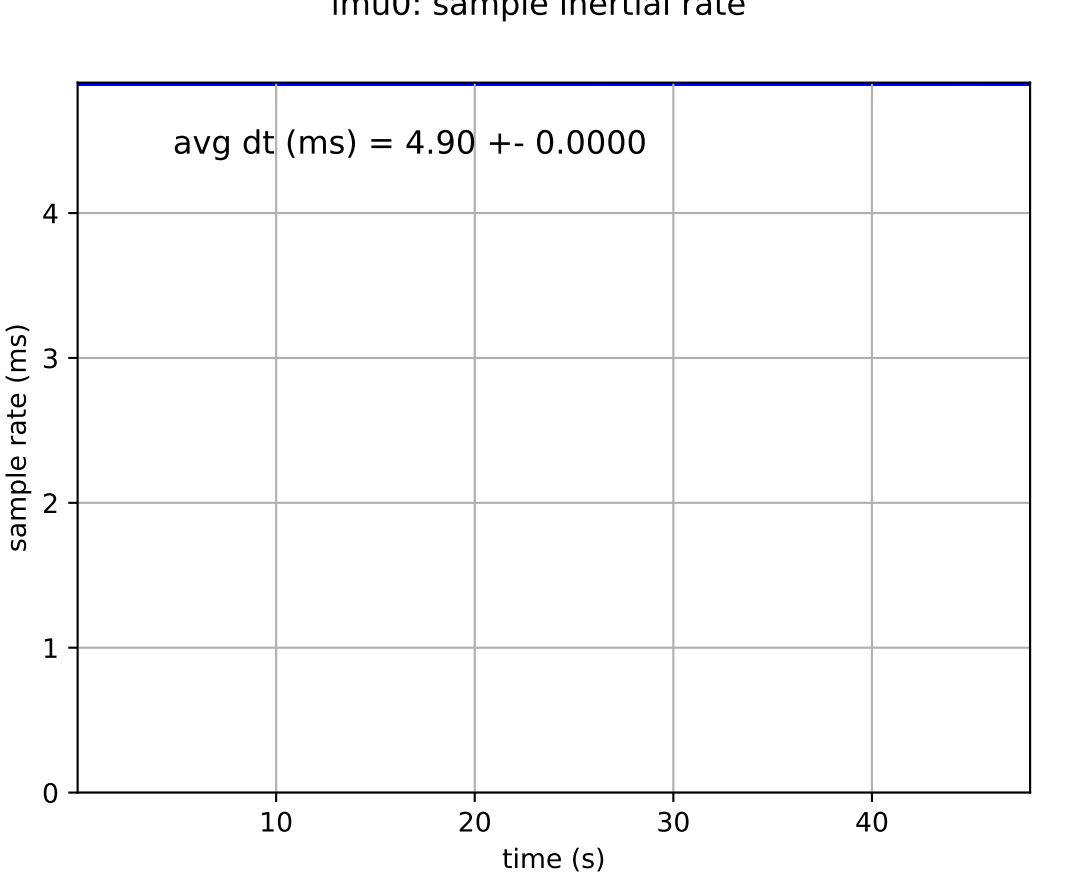
IMU configuration
==========

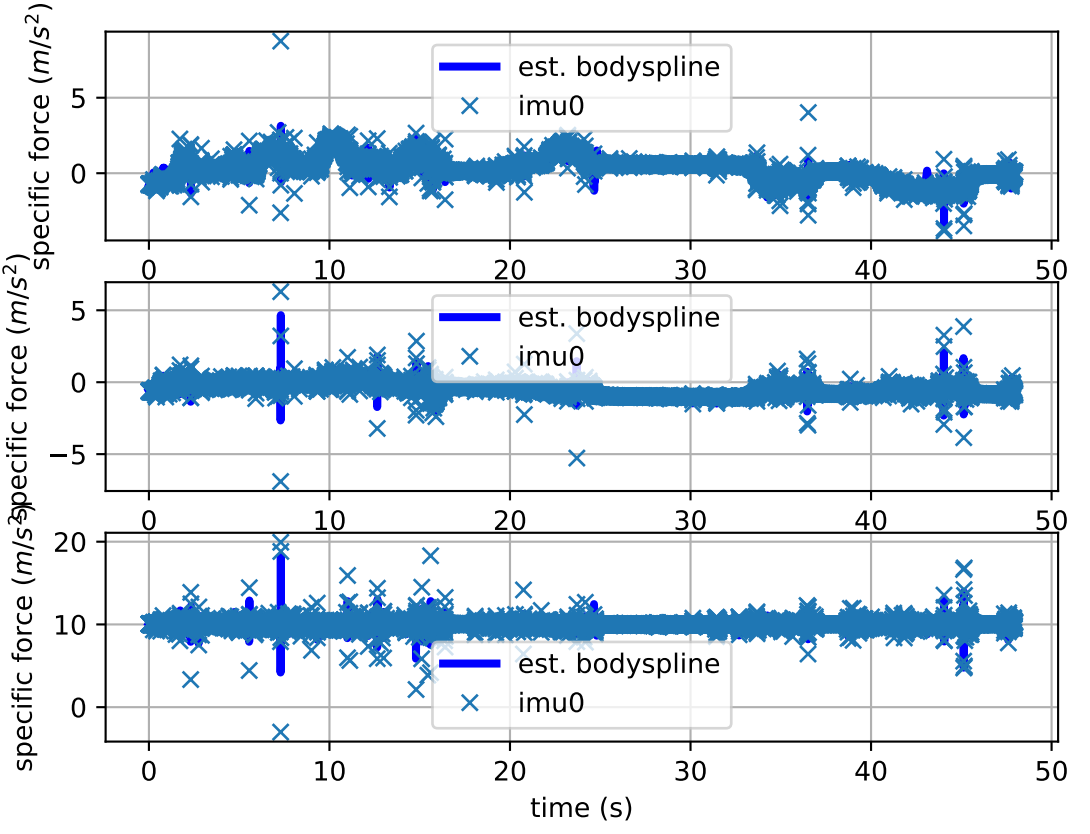
IMU0:

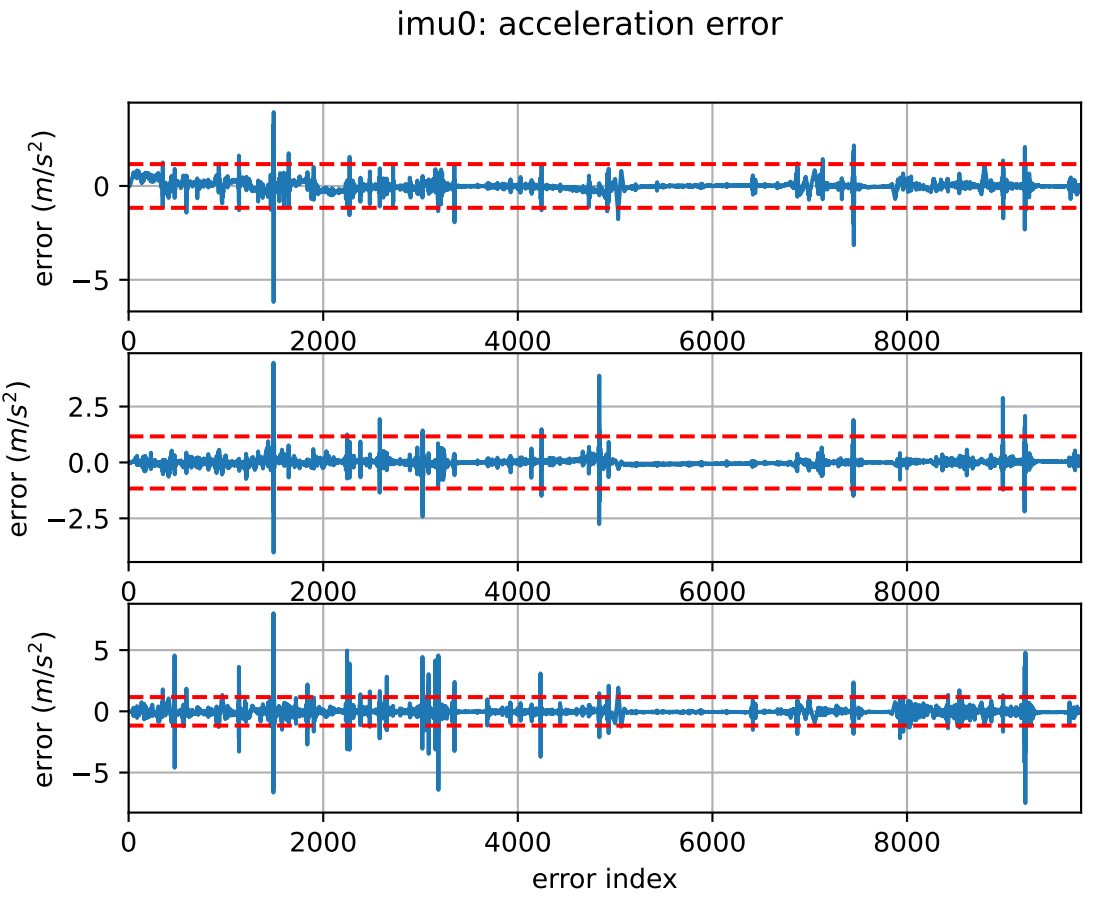
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

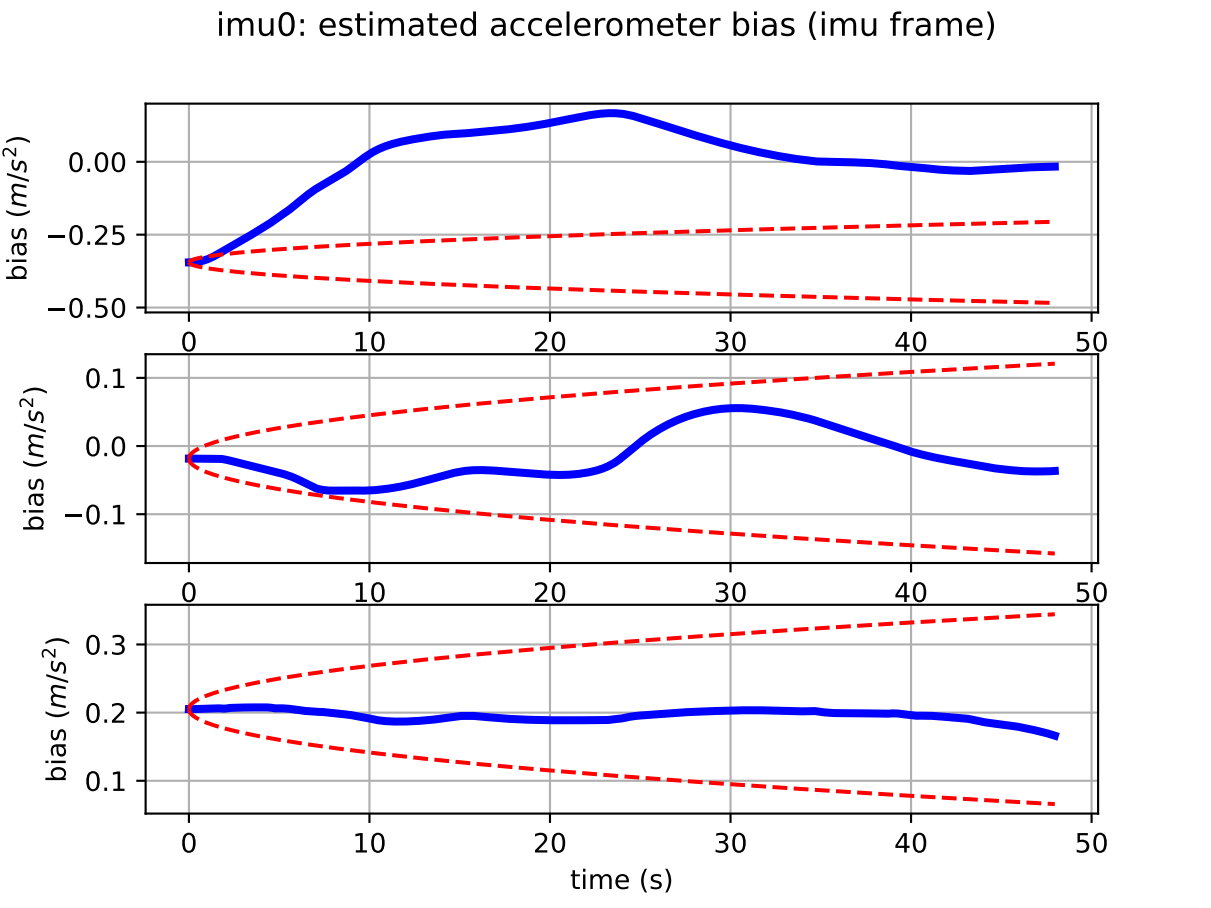
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

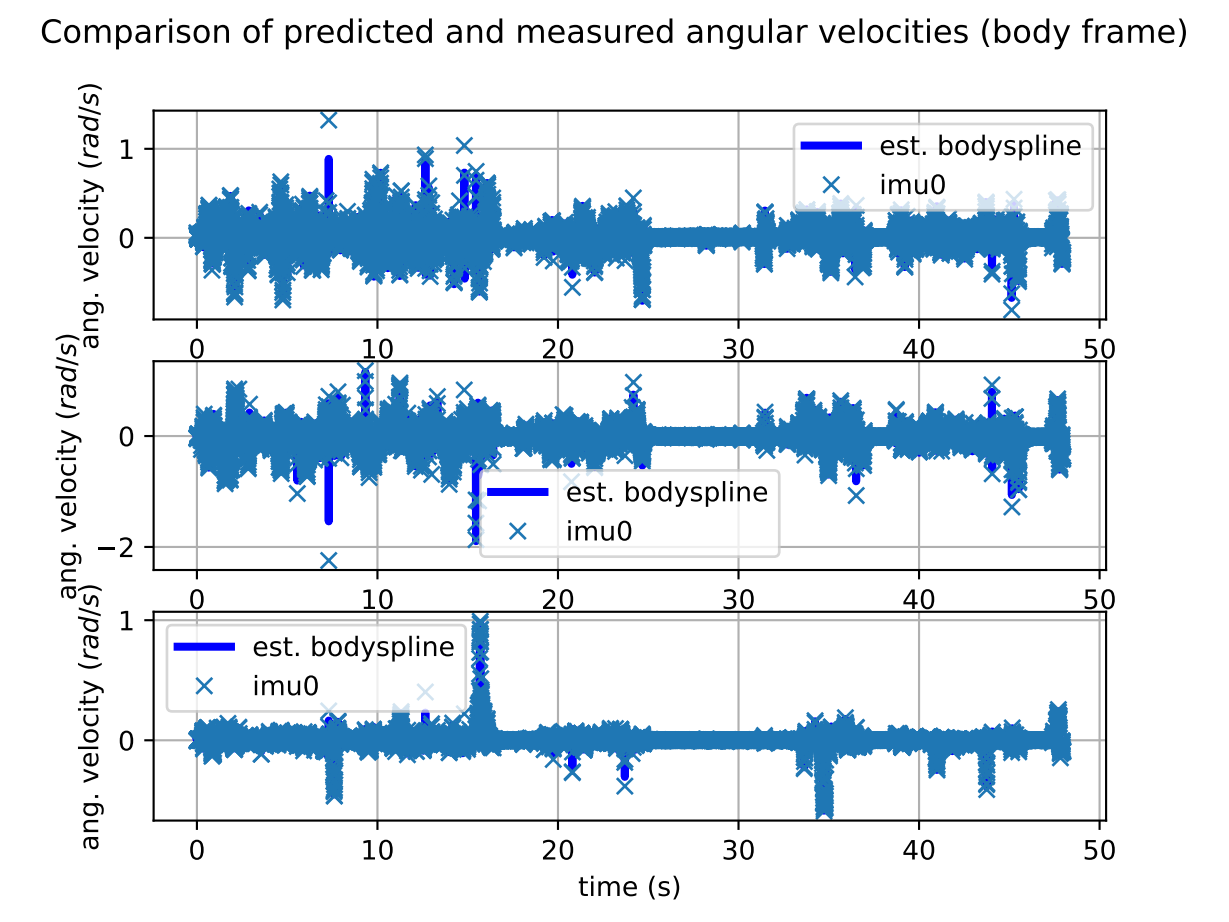

imu0: estimated poses
0.325
0.300
0.275
0.250
0.225
0.200
0.175
0.150
0.00
−0.05
−0.10
−0.15
−0.20
0.00
0.05
0.10
0.15

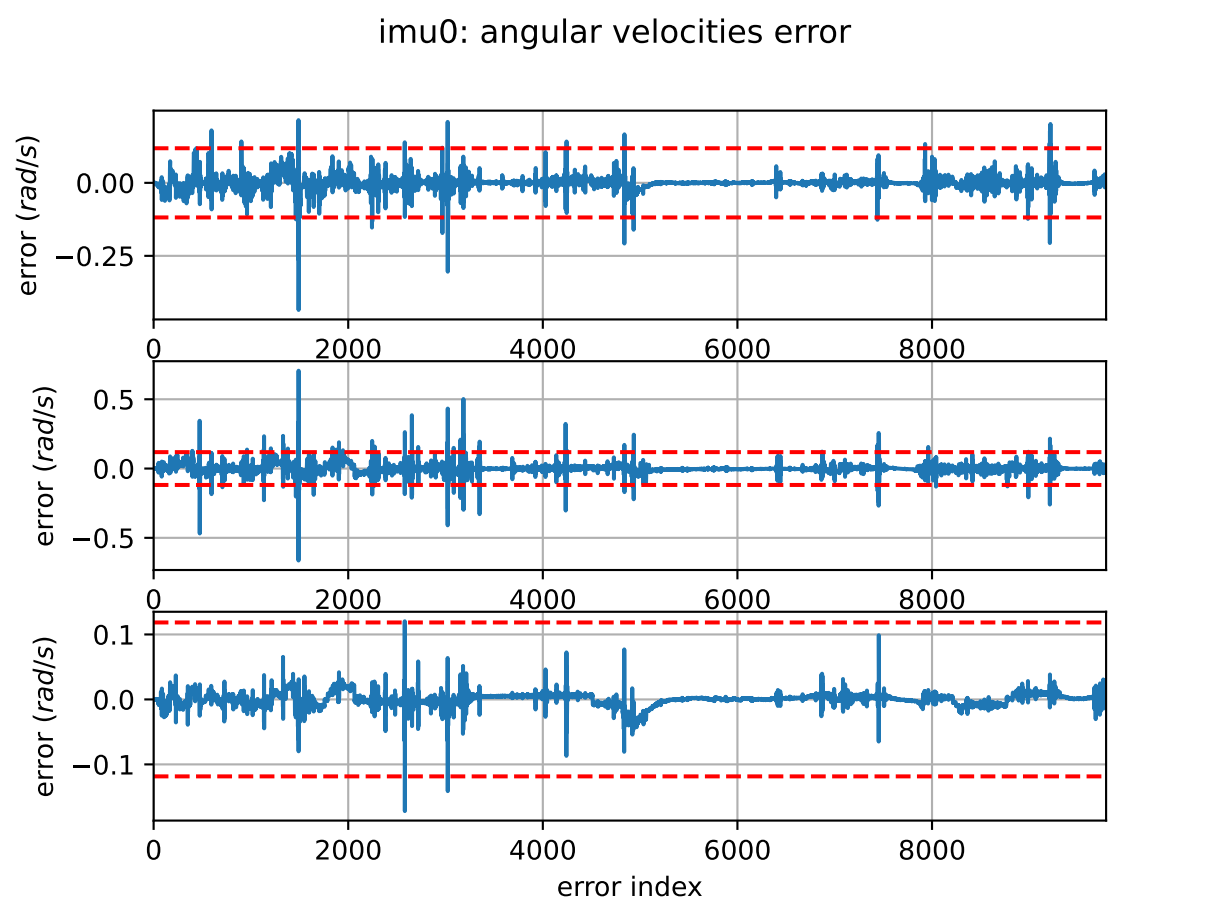

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
time (s)

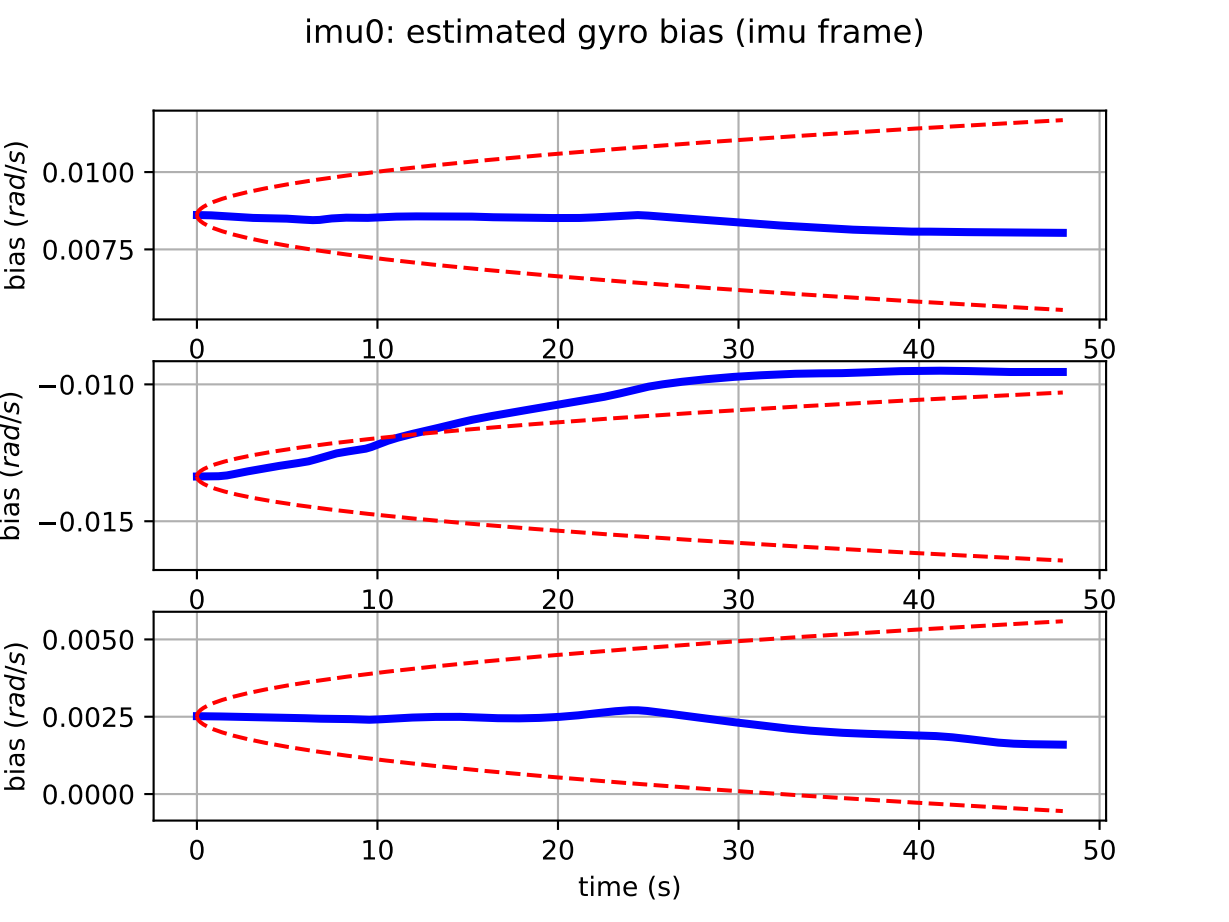

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
specific force (m/s^2)
specific force (m/s^2)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0
time (s)

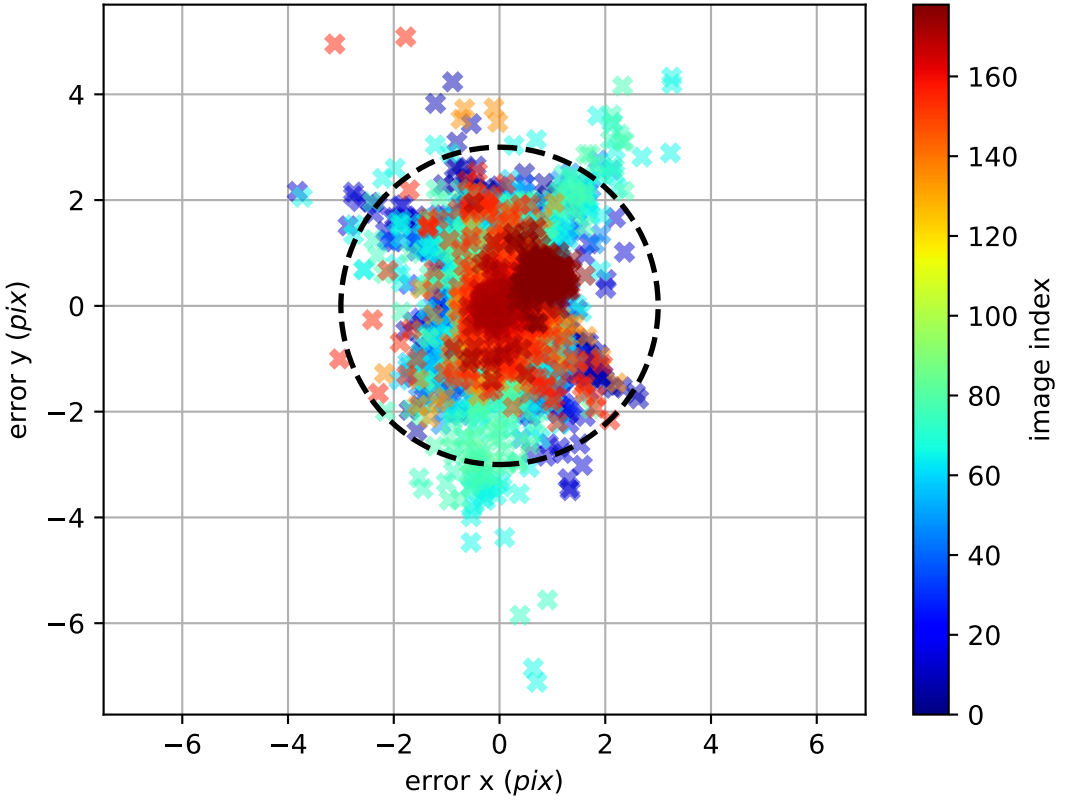

imu0: acceleration error
error (m/s^2)
0
−5
2.5
0.0
−2.5
5
0
2000
4000
6000
8000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.00
−0.25
−0.50
0
10
20
30
40
50
bias (m/s^2)
0.1
0.0
−0.1
0
10
20
30
40
50
bias (m/s^2)
0.3
0.2
0.1
0
10
20
30
40
50
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
time (s)


imu0: angular velocities error
error (rad/s)
0.00
−0.25
0
2000
4000
6000
8000
error (rad/s)
0.5
0.0
−0.5
0
2000
4000
6000
8000
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0100
0.0075
0
10
20
30
40
50
bias (rad/s)
−0.010
−0.015
0
10
20
30
40
50
bias (rad/s)
0.0050
0.0025
0.0000
0
10
20
30
40
50
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−6
−4
−2
0
2
4
6
0
20
40
60
80
100
120
140
160